

A SHARPER NUMERICAL CERTIFICATE FOR THE MATRIX MULTIPLICATION EXPONENT

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ABSTRACT. We report a new computer-assisted world-record candidate for the exponent of square matrix multiplication,

$$\omega < 2.371310.$$

The previous best published bound was $\omega < 2.371339$, obtained from the fourth power of the Coppersmith–Winograd tensor by the *More Asymmetry* analysis of Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou. We do not change their combinatorial degeneration. Instead, we re-optimize 126 term-specific six-region probability vectors that are already independent variables in their constituent-stage theorem. Replacing every nonsmooth minimum by an exact epigraph gives 42 explicit candidate inequalities and exposes a better balanced feasible point. A fresh evaluation gives the tightened exponent 2.3713098963342496; a conservative maximum-entropy dual audit gives 2.3713098963360215. We supply a standalone certificate with stored exponent 2.3713099046652784, inward inequality slack, an independent NumPy verifier, and a second term-by-term reconstruction. Finally, tensoring the fourth-power degeneration with itself gives an exact eighth-power realisation with exponent 2.3713098963357147. The complete certificate and audit trail are included as ancillary files. The analytic reduction is exact; the supplied validation is a reproducible, dual-safe binary64 computation rather than an outward-rounded interval proof.

1. INTRODUCTION

Let ω denote the infimum of the real numbers τ such that two $n \times n$ matrices over a field can be multiplied using $O(n^{\tau+\varepsilon})$ arithmetic operations for every $\varepsilon > 0$. Strassen’s seven-multiplication algorithm showed in 1969 that $\omega < \log_2 7$ [Str69]. The subsequent record sequence has been driven by increasingly refined analyses of powers of the Coppersmith–Winograd tensor [CW90]. The most recent steps are shown in Table 1.

As of August 9, 2026, the best published bound, $\omega < 2.371339$, follows from the *More Asymmetry* framework [ADV⁺25]. That work analyses $CW_5^{\otimes 4}$ and introduces six possible permutations of the roles of the X , Y , and Z variables. At every interior constituent term, the theorem permits an independent probability vector describing how much of that term is assigned to each of the six regions. The released implementation registers these vectors independently, but the resulting optimisation is large, nonsmooth, and numerically delicate. The authors report that a typical solve takes about a week and that the full eighth-power programme was infeasible with their solver.

Our contribution is deliberately narrow but consequential: we freeze the published global distributions, conditional split laws, complete split distributions, and maximum-entropy witnesses, and re-optimize only the 126 exposed constituent-stage region vectors. This is enough to improve the exact stored record point from 2.3713389005434182 to the safe value 2.3713099046652784. The rounded improvement is approximately 2.90×10^{-5} .

The main result is as follows.

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TABLE 1. Recent improvements to the square matrix multiplication exponent. The last row is the conservative decimal statement supported by the certificate in this paper.

Work	Main mechanism	Bound on ω
Alman–Vassilevska Williams [AV24]	refined laser method	2.3728596
Duan–Wu–Zhou [DWZ23]	asymmetric hashing, eighth power	2.371866
Vassilevska Williams et al. [VXXZ24]	complete split distributions	2.371552
Alman et al. [ADV ⁺ 25]	six-region More Asymmetry analysis	2.371339
This paper	exact epigraph rebalancing and certification	2.371310

Theorem 1.1 (Record-candidate bound). *The square matrix multiplication exponent satisfies*

$$\omega < 2.371310.$$

Three features of the computation are worth separating.

- (1) *The structural theorem is unchanged.* Every modified quantity is a six-entry simplex variable explicitly allowed by the constituent-stage theorem of [ADV⁺25].
- (2) *The discovery solve is not the certificate.* We use an exact epigraph formulation to find the row point, then discard its auxiliary variables and reconstruct every retained-copy and matrix-size quantity in a fresh model with explicit inward slack.
- (3) *The eighth-power statement is exact.* Tensoring the certified fourth-power degeneration with itself gives a degeneration of $\text{CW}_5^{\otimes 8}$ with the same exponent. No conjectural correlated eighth-power improvement is needed for Theorem 1.1.

The remainder of the paper gives enough of the framework to identify the modified variables, states the finite optimisation problem, and records three independent certification paths. Full parameter arrays are left to the machine-readable ancillary files rather than printed in the paper.

2. TENSOR BACKGROUND AND THE MORE ASYMMETRY PROGRAMME

2.1. Matrix multiplication tensors. For positive integers a, b, c , write

$$\langle a, b, c \rangle = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^c x_{ij} y_{jk} z_{ki}$$

for the matrix multiplication tensor. We write $\tilde{R}(T)$ for the asymptotic rank of a tensor T . Schönhage’s asymptotic sum inequality [Sch81] converts a degeneration of a low-rank tensor into a large direct sum of matrix multiplication tensors into an upper bound on ω .

The Coppersmith–Winograd tensor is

$$\begin{aligned} \text{CW}_q &= x_0 y_0 z_{q+1} + x_0 y_{q+1} z_0 + x_{q+1} y_0 z_0 \\ &\quad + \sum_{i=1}^q (x_0 y_i z_i + x_i y_0 z_i + x_i y_i z_0). \end{aligned}$$

It has border rank and asymptotic rank $q + 2$ [CW90]. The recent record bounds take $q = 5$, recursively zero out powers of CW_5 into direct sums of matrix multiplication tensors, and apply the asymptotic sum inequality.

2.2. The finite bound at the fourth power. We use the notation of [ADV⁺25, Sections 5–7]. At recursion level $\ell^* = 3$, the input tensor is $\text{CW}_5^{\otimes 2^{\ell^* - 1}} = \text{CW}_5^{\otimes 4}$. The global stage applies Theorem 5.3 of that paper, and the constituent stages apply its Theorems 6.2 and 6.4. The result is a degeneration of many copies of $\text{CW}_5^{\otimes 4n}$ into a direct sum of matrix multiplication tensors. After taking logarithms, the sufficient condition in [ADV⁺25, Equation (7)] has the form

$$\mathcal{R}(\theta) + \tau \mathcal{M}(\theta) \geq 4 \log 7. \quad (1)$$

Here θ denotes all probability distributions and complete-split data, $\mathcal{R}$ is the total logarithmic retained-copy contribution, and $\mathcal{M}$ is the logarithm of the side length of the final square matrix multiplication tensor. Any feasible parameter point satisfying Equation (1) implies $\omega \leq \tau$.

For the square problem, the six global regions have equal mass $1/6$. At every interior constituent term t , however, Theorem 6.4 permits its own vector

$$\rho_t = (\rho_{t,1}, \dots, \rho_{t,6}) \in \Delta_6, \quad \rho_{t,r} \geq 0, \quad \sum_{r=1}^6 \rho_{t,r} = 1. \quad (2)$$

The six entries select the six permutations of the roles of X, Y, Z . The released fourth-power model contains 126 interior terms with such vectors. Their conditional split distributions remain fixed in this paper; only the proportions in Equation (2) vary.

For later use, expose the three coordinate candidates before each minimum:

- $L_{r,d}^{(3)}(\rho)$, for the six level-three region retentions;
- $L_d^{(2)}(\rho)$, for the symmetric level-two retention;
- $G_{r,d}(\rho)$, for the six global-stage retentions; and
- $M_d(\rho)$, for the three output matrix-size candidates,

where $r \in [6]$ and $d \in \{X, Y, Z\}$. These are exactly the unminimised arrays computed by the released verification model. The total retained and matrix-size quantities are

$$\mathcal{R}(\rho) = \sum_{r=1}^6 \min_d L_{r,d}^{(3)}(\rho) + \min_d L_d^{(2)}(\rho) + \sum_{r=1}^6 \min_d G_{r,d}(\rho), \quad (3)$$

$$\mathcal{M}(\rho) = \min_d M_d(\rho). \quad (4)$$

Thus the exponent certified by a structural row point is

$$\Omega(\rho) = \frac{4 \log 7 - \mathcal{R}(\rho)}{\mathcal{M}(\rho)}. \quad (5)$$

Lemma 2.1 (Legality of the optimisation axis). *Fix every parameter in the published $\text{CW}_5^{\otimes 4}$ certificate except the 126 vectors $\{\rho_t\}$. If each ρ_t satisfies Equation (2), and if all mixed complete split distributions, compatibility terms, minima, and dependent auxiliary variables are recomputed, then the resulting point is admissible in Theorems 5.3, 6.2, and 6.4 of [ADV⁺25].*

Proof. For every interior constituent term t , Theorem 6.4 assigns independent nonnegative quantities $A_{t,r}$ with $\sum_r A_{t,r} = 1$. These are precisely the entries stored as the term’s local region proportions. Changing $A_{t,r}$ repartitions occurrences of the already revealed parent term among the six hashing orientations. It does not change the parent identifier, a conditional split law, or a maximum-entropy feasible set. For fixed lower policies, the level-three, level-two, and matrix-size contributions are affine in the region weights. The positional complete split distribution is affine before the global entropy and compatibility functionals are evaluated. Recomputing those nonlinear functionals and all subsequent minima therefore evaluates the same theorem at a different admissible point. $\square$

TABLE 2. Exponent values associated with the final structural row point.

Quantity	Value
Previous stored certificate	2.3713389005434182
Direct epigraph-row evaluation	2.371309896335721
Fresh verifier, exact coordinate minima	2.3713098963342496
Maximum-entropy dual-safe evaluation	2.3713098963360215
Stored certificate with safety slack	2.3713099046652784
Conservative public statement	2.371310

This lemma is the conceptual reason the improvement does not require a new laser-method argument. The work is in locating and carefully certifying a better point of the existing finite programme.

3. EXACT EPIGRAPH REBALANCING

3.1. Removing the nonsmooth minima. Direct smooth-min approximations proved useful for exploration but obscured which coordinate branches were active. We therefore introduced 14 epigraph variables

$$e_1^{(3)}, \dots, e_6^{(3)}, \quad e^{(2)}, \quad g_1, \dots, g_6, \quad m$$

and imposed the 42 inequalities

$$e_r^{(3)} \leq L_{r,d}^{(3)}(\rho), \quad r \in [6], \quad d \in \{X, Y, Z\}, \quad (6)$$

$$e^{(2)} \leq L_d^{(2)}(\rho), \quad d \in \{X, Y, Z\}, \quad (7)$$

$$g_r \leq G_{r,d}(\rho), \quad r \in [6], \quad d \in \{X, Y, Z\}, \quad (8)$$

$$m \leq M_d(\rho), \quad d \in \{X, Y, Z\}. \quad (9)$$

At any optimum the appropriate inequalities are tight, and Equation (5) becomes

$$\min_{\rho, e, g, m} \frac{4 \log 7 - \sum_{r=1}^6 e_r^{(3)} - e^{(2)} - \sum_{r=1}^6 g_r}{m} \quad (10)$$

subject to Equations (2) and (6) to (9). The formulation has 756 region entries and 14 epigraph variables. We parameterised each supported simplex by a masked softmax and supplied analytic derivatives of every objective and constraint.

The optimisation remains nonconvex: in particular, the global complete-split entropy depends nonlinearly on the mixed region rows. Global optimality is neither claimed nor needed. Any point returned by the search is evaluated independently, and a single feasible improvement suffices.

3.2. Search and reconstruction. Starting from the published certificate, we first used smooth continuation to approach the active-set intersection. We then solved Equation (10) by sequential quadratic programming, using the SLSQP implementation in SciPy [Kra88, VGO⁺20]. The final scaled active-set stage converged in two iterations; the minimum epigraph residual was zero at displayed precision, and the epigraph and direct objectives agreed to 3×10^{-15} .

The solver’s auxiliary variables are not trusted as a certificate. We retain only the 126 row vectors, install them into a fresh copy of the released model, recompute every complete split distribution and unminimised candidate, take the exact coordinate minima, and rebuild the retained-copy, matrix-size, and exponent variables with inward slack. The numerical progression is summarised in Table 2.

The change audit finds 543 modified scalar entries among the 24,855 parameters: 528 coordinates in the 126 legal region rows and 15 reconstructed auxiliary coordinates. No global distribution, split law, complete split distribution, or entropy witness is changed.

4. CERTIFICATION

4.1. A dual-safe entropy calculation. The More Asymmetry programme contains many penalties

$$P_\alpha = \max\{H(p) : Ap = \mu, p \geq 0\} - H(\alpha), \quad H(p) = - \sum_s p_s \log p_s.$$

The released certificate stores both an approximate maximum-entropy primal distribution and KKT multipliers. We remove any logical dependence on the approximate KKT equalities by using weak duality.

Lemma 4.1 (Entropy dual bound). *For every vector λ and every feasible marginal vector μ ,*

$$\max_{p \geq 0, Ap = \mu} H(p) \leq D(\lambda; \mu) := \sum_s \exp((A^\top \lambda)_s - 1) - \lambda^\top \mu. \quad (11)$$

Proof. Use the Lagrangian $H(p) + \lambda^\top (Ap - \mu)$ and the scalar identity

$$\sup_{x \geq 0} \{-x \log x + yx\} = e^{y-1}.$$

Taking the supremum over all entries of p gives Equation (11); weak duality completes the proof. $\square$

For every one of the 762 maximum-entropy subproblems, we evaluate D at the stored multipliers and replace the primal entropy by this upper bound. This can only increase P_α , decrease the relevant retention candidate, and hence increase the inferred exponent. The resulting conservative value is

$$\omega_{\text{dual}} = 2.3713098963360215,$$

which remains 1.0366×10^{-7} below the stated decimal bound. The approximate KKT residuals are therefore diagnostics, not proof assumptions. The dual-safe logarithmic rates are

$$\mathcal{M} = 2.0967467985574295, \quad \mathcal{R}_{\text{dual}} = 2.8116041626911494,$$

so that $\omega_{\text{dual}} = (4 \log 7 - \mathcal{R}_{\text{dual}})/\mathcal{M}$.

4.2. Independent evaluations. We performed three complementary checks.

- (1) A fresh value-only NumPy port reconstructs the 24,855-parameter model from the MAT file. It does not use SNOPT or the released forward automatic-differentiation class.
- (2) A second dense checker explicitly drives every term's initialisation, downward mass propagation, and upward contribution propagation, then independently sums every unminimised level-three, level-two, global, and matrix candidate. It does not call the aggregate objective routine.
- (3) A complete parameter-difference audit compares the new MAT file with the published source certificate group by group and confirms that every structural change lies in one of the 126 permitted region simplexes.

The safe certificate has SHA-256 digest

65d78064ef3686f850558b3881446403afe9d4910d2d2e046c3dab92c390ff83.

TABLE 3. Numerical certification summary. Inequality residuals use the convention that positive values violate feasibility.

Check	Result
Parameters reconstructed	24,855
Maximum inequality residual	$-9.9999997 \times 10^{-10}$
Schönhage inequality residual	$-2.0967486 \times 10^{-9}$
Maximum bound violation	0
Maximum optimised-row simplex residual	2.22×10^{-16}
Maximum structural simplex residual inherited from source	8.15×10^{-13}
Maximum component discrepancy between independent evaluators	3.31×10^{-12}
Difference between fourth- and eighth-power evaluations	1.47×10^{-12}
Unexpected changed parameter groups	0

At its stored exponent, the two sides of the certificate inequality are

$$7.783640598318 \quad \text{and} \quad 4 \log 7 = 7.783640596221,$$

respectively. The dual-safe reconstruction has logarithmic margin 1.7464×10^{-8} at the stored exponent.

Proof of Theorem 1.1. By Lemma 2.1, the 126 optimised row vectors are admissible parameters in the degeneration theorems of [ADV⁺25]. The fresh reconstruction evaluates all dependent complete-split and compatibility quantities. By Lemma 4.1, the dual entropy audit bounds every maximum-entropy penalty in the conservative direction and yields $\Omega(\rho) \leq 2.3713098963360215 < 2.371310$. Substitution into Equation (1) and Schönhage’s asymptotic sum inequality therefore imply $\omega < 2.371310$. $\square$

Remark 4.2 (Floating-point status). The ancillary verification follows the numerical-certificate convention of the recent matrix multiplication literature: it supplies a finite floating-point witness, independent evaluators, and a headline rounded away from the computed value. The entropy dual removes the only use of approximate KKT optimality. A fully formal interval-arithmetic replay would be a useful final layer; it is not claimed here. Until the certificate has also been reproduced externally, we regard the result as a world-record candidate rather than an independently established published record.

5. THE EXACT EIGHTH-POWER LIFT

The new row point is found in the fourth-power programme, but the requested eighth-power realisation is exact and immediate.

Proposition 5.1 (Tensor-square lift). *Suppose a fourth-power degeneration satisfies Equation (1) for an exponent τ . Tensoring that degeneration with itself gives an eighth-power degeneration satisfying the corresponding eighth-power inequality for the same τ .*

Proof. For large n , suppose the fourth-power construction degenerates a direct sum of $K_n = 2^{o(n)}$ copies of $\text{CW}_5^{\otimes 4n}$ into a direct sum of V_n copies of $\langle A_n, B_n, C_n \rangle$. Tensoring the entire degeneration with itself gives a degeneration of a direct sum of $K_n^2 = 2^{o(n)}$ copies of $\text{CW}_5^{\otimes 8n}$ into V_n^2 copies of $\langle A_n^2, B_n^2, C_n^2 \rangle$. Consequently the logarithmic retained-copy contribution and every logarithmic matrix-size contribution double, while the asymptotic-rank target changes from $4 \log 7$ to $8 \log 7$. The sufficient inequality is therefore precisely the square of the fourth-power inequality. $\square$

As an independent numerical check, we also evaluated the product construction in the 36 ordered pairs of the six hashing orientations. The result is

$$\omega_{\text{product}}^{(8)} = 2.3713098963357147,$$

only 1.47×10^{-12} above the fresh fourth-power reconstruction. The maximum row and column marginal residual is 3.63×10^{-14} . An aggregate 36-cell evaluator gives 2.3713098963357138. These computations are not needed for Proposition 5.1, but they check the eighth-power bookkeeping independently.

We additionally searched fixed-marginal orientation couplings and counterpart-specialised policies for the 80 highest-mass refined parents. After rebasing on the final epigraph point, none improved the tensor product. Thus the present exponent improvement is a genuine fourth-power rebalancing inherited exactly by the eighth power, not a claimed new correlated eighth-power construction.

6. DISCUSSION

The improvement is numerically small but structurally informative. The published programme already contained the necessary freedom; what remained was a difficult nonsmooth balancing problem. The coordinate minima in Equations (3) and (4) create narrow active-set intersections, so a smooth surrogate or a generic large-scale SQP solve can stop near, but not at, the balance that controls the true objective. The explicit 42-inequality epigraph made those branches visible and allowed a targeted final solve.

There are two immediate directions for further work. First, one can repeat the active-set strategy while releasing additional split distributions, rather than only the region weights. Second, the full $\text{CW}_5^{\otimes 8}$ programme permits non-product choices not represented by the exact tensor square. Our exploratory searches found no gain on several natural restricted families, but they cover only a small part of the full eighth-power feasible set. A sparse reverse-mode implementation of the full programme, combined with exact epigraph handling and certified entropy duals, remains a plausible route to another improvement.

The broader lesson is methodological. In record computations based on entropy and nested minima, optimisation and certification should be separated. A solver proposes structural parameters; a fresh epigraph evaluation identifies the exact active branches; weak duality supplies conservative entropy bounds; and an independent programme reconstructs the theorem’s finite inequality. This workflow is cheap relative to the original parameter search and materially reduces the risk that a nominal improvement is an artefact of smoothing, stale auxiliaries, or solver tolerances.

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7. REPRODUCIBILITY AND ANCILLARY FILES

The principal files in the ancillary archive are listed in Table 4.

The authors’ public archive contains the base implementation and published 2.371339 certificate [ADV⁺24]. The principal checks are:

TABLE 4. Principal machine-readable files supplied with the manuscript.

File	Purpose
W1.00_2.371310_safe_final.mat	Standalone 24,855-parameter safe certificate.
p4_region_epigraph_stage4.npz	The 126 optimised region rows and epigraph data.
python_verifier.py	Independent value-only reconstruction of the released model.
dual_safe_verifier.py	Entropy-dual conservative audit.
p4_dense_epigraph_final_independent_verification.json	Term-by-term dense audit.
refined_aggregate_product_epigraph_final.json	Explicit eighth-power product evaluation.
SHA256SUMS	Cryptographic digests for the archive.

```
python3 python_verifier.py W1.00_2.371310_safe_final.mat --top 20
python3 dual_safe_verifier.py W1.00_2.371310_safe_final.mat
```

The expected headline values are those in Tables 2 and 3.

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